

Written Justification

System: Student Project Portfolio & Showcase App

Architecture Selection

We chose **Microservices Architecture** for the *Student Project Portfolio & Showcase App*.

Architectural Choice

The selected architectural style is Microservices Architecture. The main functions of the system can be separated into services such as project showcase, project submission, judging, voter verification and voting, leaderboard, and voting-window management. Each service can handle its own responsibility while communicating with the other services through defined interfaces.

Reason 1

The system can receive many votes during the project expo, so the voting and leaderboard parts may have more load than other parts of the application. With microservices, these services can be scaled separately without scaling the complete application.

Reason 2

The application has different types of work, such as submitting projects, judging them, verifying voters, and managing the voting period. Keeping these functions as separate services makes the system easier to maintain and also reduces the effect of a problem in one service on the other parts.

Security Advantage

Voter verification can be handled separately before a vote is accepted. The voting service can check the verified identity and make sure that the same user does not vote more than once for a project. This gives a clear place to apply the OTP verification and duplicate or Sybil-voting checks required by the system.

Performance Benefit

The voting and real-time leaderboard services can be scaled independently when the number of users increases. This is useful because the system requires the leaderboard and voting updates to respond in real time, with the API response latency targeted below 300 ms.